

Experimental apparatus for universal spin-motion control in multi-ion registers



Donovan Webb
St Peter's College
University of Oxford

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Towards Hybrid System Control and Quantum Error Mitigation

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Having established and characterised the experimental apparatus and demonstrated its application to *k-copy* randomised benchmarking, this chapter considers future research goals utilising the multi-qubit arrays enabled through the *NPL* trap (§ ??), the single addressing system (§ ??), and the developed control over both the discrete qubits, as well as the continuous motional modes.

In § 1.1 we establish a set of unitary operations required for universal continuous-variable (CV) control of the ion motion and discuss the extensions to hybrid spin-oscillator systems. Following on from recent progress in spin-conditioned bosonic operations demonstrated within our group [1–5], we are interested in

developing the required addressing capabilities to scale to multiple qubits and motional modes. We outline the method we use to generate such spin-oscillator interactions experimentally, and demonstrate the generation of a squeezed state on the new apparatus. We then outline the next steps required for demonstrating the remaining single- and two-mode interactions, and the possible experiments this will enable.

The second half of this chapter considers an error mitigation technique applicable to both discrete and continuous systems. Developing this, and associated methods on our trapped ion platform will enable substantial reductions of certain errors, with significantly less resources required than full error correction. We give a short introduction to the purpose of quantum error mitigation, followed by an outline of virtual distillation [6–8] — the specific method considered here. We provide a demonstration of this protocol applied to a discrete-variable only system. Finally, we explore the impact of structured error channels on the efficacy of virtual distillation, particularly focussing on correlated noise due to a common fluctuating control field as discussed in the chapter ?? § ??. The *k-copy* protocol developed in that chapter is utilised to quantify this unfavourable error, and inform predictions on the performance of virtual distillation in this context. The section is concluded with a discussion on the application of virtual distillation to larger system sizes, and to hybrid systems, presenting a scalable method within reach of our apparatus.

1.1 Hybrid Spin-Oscillator Control in Trapped Ions

CV quantum computation [9] offers an alternative strategy for encoding quantum information, utilising quantum harmonic oscillators, termed *qumodes*, as the fundamental quantum object, rather than discrete variable (DV) systems, e.g. the two-level qubit. A single ideal qumode possesses an infinite dimensional Hilbert space, allowing for high (in this case arbitrarily high) density encoding of information in physical information carriers. This opens an alternative avenue for scaling QIPs, requiring dramatically less physical carriers compared to qubit-only systems. The

inclusion of both CV and DV subsystems, termed the hybrid model [10], retains the convenient control and readout of discrete qubits while providing the larger state space and native interactions available to qumodes. For example, reference [11] considers a factoring algorithm with the required physical resources independent of the number to be factored. Remarkably, this scheme requires only a single qubit and three oscillators, however requires prohibitively large energy states of the oscillator. Experimentally, we will not be able to realise arbitrarily high energy oscillator states, but it is still an open question of what information density is possible in accessible engineered quantum oscillator.

Hybrid spin-oscillator computation has been studied in the context of superconducting cavity-QED [12–16], neutral atom in optical tweezers [17–19], and in the trapped-ion platform [20–22]. We shall focus on the trapped-ion implementation, where the DV system is the collection of qubits embedded in the electron energy levels, and the qumodes are the shared secular motional modes of the ion crystal within the trapping pseudopotential.

Trapped ions benefit from multiple native methods to drive, and couple qubit and qumode: The qumodes may be directly driven through parametric trap potential modulation [23–25], or through near mode-resonance RF fields; the qubit may be driven either by optical fields (in the case of optical qubit encoding), or through microwaves (in the case of ground state encoding) resonant with the energy-state transition; the qubit and qumode may be coupled through qubit-state dependent potential shifts [26, 27], or through optical fields tuned near to motional-mode frequency sidebands of the qubit-carrier transition, for example through the Jaynes-Cummings interaction (§ ??), or the Mølmer-Sørensen (MS) SDF [28]. Taken together, these native control and coupling mechanisms, combined with the long coherence times of both the qubit and qumode, make the trapped-ion platform well-suited for exploring hybrid CV–DV quantum computation.

The apparatus developed in this thesis has demonstrated good qubit and radial motional coherence times, provides access to multiple qubits and qumodes, and includes

Interaction name	Oscillator $\hat{H}_{\text{osc}}$	Gauss.	N_{osc}	Two-SDF method		
				m	$\hat{\sigma}$	Ω_{eff}
Displacement	$\hat{a}e^{i\phi} + \hat{a}^\dagger e^{-i\phi}$	✓	1	-	$\hat{\sigma}_\alpha$	Ω_α
Parity	$(2\hat{a}^\dagger\hat{a} + 1)\sin(\phi)$	✓	1	+1	$\hat{\sigma}_\gamma$	Ω_2
Beam splitter	$\hat{a}\hat{b}^\dagger e^{i\phi} + \hat{a}^\dagger\hat{b}e^{-i\phi}$	✓	2	+1	$\hat{\sigma}_\gamma$	Ω_2
Squeezing	$\hat{a}^2 e^{i\phi} + \hat{a}^{\dagger 2} e^{-i\phi}$	✓	1	-1	$\hat{\sigma}_\gamma$	Ω_2
Trisqueezing	$\hat{a}^3 e^{i\phi} + \hat{a}^{\dagger 3} e^{-i\phi}$	✗	1	-2	$\hat{\sigma}_\beta$	Ω_3

Table 1.1: A suggested set of unitary interactions to compose a universal gate set for continuous variable computation. The name for the CV interaction, the bosonic operator component of the Hamiltonian, if it is Gaussian, and the number of oscillators it acts on are given. These CV interactions are accessible and may be extended to hybrid-interactions (CV-DV) by using the protocol presented in reference [20], here termed the “Two-SDF” method. The required relative detuning setting, m , the resulting spin conditioning, $\hat{\sigma}$, and the effective coupling strength Ω_{eff} are shown here. The first row is a single SDF with strength Ω_α . The other coupling strengths are found via the 2nd and 3rd terms of the Magnus expansion: $\Omega_2 = \Omega_\alpha \Omega_{\alpha'}/\Delta$, $\Omega_3 = \Omega_\alpha^2 \Omega_{\alpha'}/\Delta^2$.

a single-addressing system for selectively driving their coupling. These capabilities make it well-placed for studying methods for scaling hybrid-system control.

1.1.1 Theory

Gateset for a Hybrid System

For any computing paradigm, we must establish a set of interactions required for universality. In DV-only systems, quantum universality may be realised by a finite set of Clifford operations plus one non-Clifford gate [29, 30]. CV-only universality can analogously be realised by a finite set of Gaussian operations plus one non-Gaussian gate [9, 31]. The term Gaussian is understood in the position-momentum representation of the qumode, where the bare oscillator Hamiltonian is given by,

$$\hat{H}_{\text{osc},0} = \hbar\omega_{\text{osc}}(\hat{x}^2 + \hat{p}^2), \quad (1.1)$$

with quadratures,

$$\hat{x} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger), \quad (1.2)$$

$$\hat{p} = \frac{-i}{2}(\hat{a} - \hat{a}^\dagger), \quad (1.3)$$

where $\hat{a}^\dagger$, $\hat{a}$ are the creation and annihilation ladder operators of the mode with frequency, ω_{osc} . Gaussian operators are then the set of interactions containing at most quadratic terms in $\hat{x}$ and $\hat{p}$, therefore mapping Gaussian input states to Gaussian output states.

The condition for universality of a hybrid CV-DV is somewhat ambiguous as both the CV and DV subsystems are themselves sufficient for universality, however this presents an opportunity: the freedom in selecting the chosen gate-set for universality, combining both Clifford, non-Clifford, Gaussian, non-Gaussian, and spin-oscillator interactions together, allows for selecting efficient, native, and simply accessed interactions of the physical platform. We consider the over-complete instruction set presented in Table 1.1, consisting of spin-conditioned Gaussian and non-Gaussian interactions. This set of interactions on the bosonic subsystems are themselves universal, but the spin-conditioning provides a bridge between CV and DV subsystems, allowing for readout and control to be implemented on ancilla qubits. Specifically, we consider a cubic CV instruction set [9, 12, 32], taking the Gaussian operations and supplementing with a non-Gaussian “tri-squeezing” gate, with Hamiltonian,

$$\hat{H}_T \propto \hat{a}^3 + \hat{a}^{\dagger 3}. \quad (1.4)$$

Spin-conditioned tri-squeezing has been shown previously in our group [1], utilising the protocol given in reference [20], here termed the “two-SDF” method.

The remaining non-linear Gaussian operations; squeezing, parity, and beam-splitter, may be generated through the same “two-SDF” mechanism, which we shall detail in the following section, and have also been previously demonstrated [1, 3, 4, 33].

The displacement operation can be realised either spin-independently by RF fields near resonance to the motional-mode frequencies (§ ??), or spin-conditioned using various SDF implementations, such as by the MS method using bichromatic fields (§ ??), or by static Stark-shift gradients due to an optical tweezer [27].

As highlighted in this section, the required unitaries for universal hybrid CV-DV control may all be accessed by a common protocol: The “two-SDF” method. This greatly simplifies experimental hardware and control complexity. We shall give an overview of this method in the following section, and provide a practical route and demonstration utilising four optical frequencies with fixed relative detunings and phases.

- Do we need to define each interaction, sqz, parity, bs, currently only in supplemental?
- Could give flow fields for single-mode operators (could be a nice figure, and I like that representation).

Gate Implementation

To implement the unitaries given in table 1.1 with spin-dependent control, we use the “two-SDF” method proposed in reference [20]. The non-linear bosonic interactions are effectively driven through applying two simultaneous linear spin-dependent forces with non-commuting spin conditioning, on a single qubit. The two simultaneous SDFs, with Hamiltonians and parameters labelled by subscripts α , and β , are applied off-resonantly relative to motional modes with creation operators $\hat{a}_\alpha^\dagger$ and $\hat{a}_\beta^\dagger$, respectively. The resulting Hamiltonian is given generally by,

$$\hat{H}_{2\text{SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (1.5)$$

$$= \frac{\hbar\Omega_\alpha}{2}\hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (1.6)$$

$$+ \frac{\hbar\Omega_\beta}{2}\hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right), \quad (1.7)$$

where the individual SDF strengths are Ω_i , the detuning from each mode is Δ_i , the motional phase is ϕ_i , and we ensure the spin-conditioning anticommute $\{\hat{\sigma}_\alpha, \hat{\sigma}_\beta\} = 0$, $[\hat{\sigma}_\alpha, \hat{\sigma}_\beta] = 2i\epsilon_{\alpha\beta\gamma}\hat{\sigma}_\gamma$, where $\epsilon_{\alpha\beta\gamma}$ is the Levi-Civita symbols. The unitary dynamics of $\hat{H}_{2\text{SDF}}$ may be evaluated by the Magnus expansion [34], with details given in references [3, 4, 20], and in appendix A.8. To lend some intuition on the resulting

physics, we highlight the main ideas here: The Magnus expansion expresses the generator of the unitary evolution as an infinite series of terms, with successive terms proportional to increasingly higher-order nested commutators of the Hamiltonian evaluated at different times. Due to the non-commuting spin conditioning, this series does not truncate, leading to arbitrarily high non-linear bosonic interactions effectively driven via the application of only two linear SDFs.

To utilise specific non-linear terms within this infinite series, matching the desired interactions given in table 1.1, we constrain the relative detunings of the two SDFs to $\Delta_\beta = m\Delta_\alpha$, where m is some integer given in the table. This immediately brings components of the Magnus terms matching the desired non-linear $\hat{H}_{\text{osc}}$ into resonance. Any undesired higher-order terms that are also brought into resonance can be arbitrarily suppressed by increasing the detuning, as the effective coupling of the n th-order Magnus term scales as $\Omega_{\text{eff}} \propto (\Omega_\alpha/\Delta)^{n+1}$ (relabelling $\Delta_\alpha = \Delta$ and assuming $\Omega_\alpha = \Omega_\beta$). The remaining error terms are far off-resonant, and may also be further suppressed by ensuring $\Delta \gg \Omega_\alpha\Omega_\beta$, and utilising appropriate smooth amplitude and phase envelopes for the pulses.

In this regime, the unitary dynamics of $\hat{H}_{2\text{SDF}}$ are well approximated by the evolution of time-independent effective Hamiltonians. For example, the effective squeezing interaction is driven when $m = -1$, and both SDFs act on the same mode, $\hat{a}_\alpha^\dagger = \hat{a}_\beta^\dagger = \hat{a}^\dagger$ yielding,

$$\hat{H}_{\text{eff,sqz}} = \frac{\hbar\Omega_2}{2} \epsilon_{\alpha\beta\gamma} \hat{\sigma}_\gamma (\hat{a}^2 e^{i\phi'_p} + \hat{a}^{\dagger 2} e^{-i\phi'_p}), \quad (1.8)$$

where $\Omega_2 = \frac{\Omega_\alpha\Omega_\beta}{\Delta}$ is an effective coupling strength, and phase $\phi'_p = \phi_\alpha + \phi_\beta + \frac{\pi}{2}$. The derivation and forms of the other spin-conditioned effective non-linear bosonic interactions are given in appendix A.8.

Frequency and spin-conditioning control of two SDFs therefore allow for the generation of the entire hybrid CV-DV gate-set proposed in table 1.1. After a discussion on how we can characterise the oscillator state, we shall present the experimental scheme for generating these interactions, and demonstrate the creation of a squeezed state by this method.

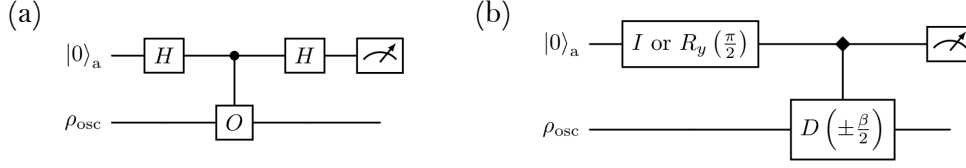


Figure 1.1: State tomography for a single mode through measurement of an ancilla qubit. (a) The Hadamard test method for measuring the expectation value of an arbitrary operator, $\hat{O}$, applied to the oscillator state, utilising an ancilliary qubit. (b) A Hadamard test circuit for measuring the characteristic function, $\chi(\beta)$, of the oscillator using native single qubit rotations and σ_x -conditioned spin-dependent forces (symmetric control represented by the diamond symbol) [35].

Readout

With the above methods we may generate interesting non-classical bosonic states, now we provide a method for characterising them.

In contrast to a two-level system, the state of an oscillator may not be directly readout through state dependent fluorescence. This is due to the much larger state-space of the qumode, and the technical difficulty in direct motional readout on the ion-trap platform. To solve these issues, a modular readout scheme is used, involving indirect readout of the oscillator state via a discrete collection of controlled unitaries subject to an ancillary qubit, and computational basis measurements of that ancilla.

The information is mapped to the ancilla by a Hadamard test [36], shown in figure 1.1 (a). Here the expectation value of the operator $\hat{O}$ on the oscillator, $\langle \hat{O} \rangle = \text{Tr}(\hat{O}\rho_{osc})$, is recovered through exclusive measurements of the ancilla qubit after qumode-dependent phase “kick-backs” are applied. The Hadamard test has the convenient relation between computational basis measurements of the ancilla, and the applied operator, $\langle \hat{\sigma}_z \rangle = \text{Re}[\text{Tr}(\hat{O}\rho_{osc})]$.

Full state tomography may now be formulated by repeat measurements of the conditional displacement operation, $\hat{O} = e^{-i\theta} \hat{D}(\beta)$, to recover the characteristic function, defined as $\chi = \text{Tr}(D(\beta)\rho_{osc})$, with β being a complex parameter with displacement magnitude, $|\beta|$, and displacement axis, $\arg(\beta)$. Final computational

state measurements of the Hadamard test circuit then yield,

$$\langle \sigma_z \rangle = \text{Re} \left[e^{-i\theta} \text{Tr}(D(\beta) \rho_{\text{osc}}) \right], \quad (1.9)$$

$$= \cos(\theta) \text{Re}[\chi] + \sin(\theta) \text{Im}[\chi], \quad (1.10)$$

where θ may be set to 0 or $\pi/2$ to recover the real or imaginary part of the characteristic function respectively. Each measurement provides only 1-bit of information on the (continuous) oscillator state, thus to approximate a useful reconstruction, many samples with a range of β values must be collected.

Once the characteristic function has been measure, it can be converted to other familiar representations of the oscillator, such as in the Fock-state basis, or Wigner representation [37–39]. In this work, we shall keep to the conveniently measured characteristic function.

To give two concrete example, the vacuum state characteristic function is the circularly symmetric Gaussian state,

$$\chi_0 = \exp \left(\frac{-|\beta|^2}{2} \right), \quad (1.11)$$

while the squeezed state, generated by application of the Hamiltonian in equation 1.8 to the vacuum state, breaks this symmetry due to a contraction on one axis, and a corresponding expansion on the other [40],

$$\chi_s = \exp \left(\frac{-|\beta|^2}{2} \left(e^{2r} \cos^2(\phi) + e^{-2r} \sin^2(\phi) \right) \right), \quad (1.12)$$

where the complex squeezing parameter is $\xi = r e^{i\phi_s} = \Omega_2 e^{i\phi'_p t}$, and ϕ is the angle between squeezing axis and probe axis, $\phi' = \arg(\beta) - \frac{\phi_s}{2}$.

Experimentally, $\hat{\sigma}_x$ conditioned SDFs are simply accessed by the bichromatic field method discussed in § ?? . The Hadamard test circuit is then decomposed to native gates, shown in figure 1.1 (b), first described in reference [35]. The proceeding I or $R_y(\pi/2)$ set the quadrature to measure; $\text{Re}(\chi)$ or $\text{Im}(\chi)$ respectively. This circuit provides a practical method for characterising a single qumode, which we shall use in the following experimental demonstration. We note, similar to the constraints of state characterisation discussed in § ?? , full tomographic reconstruction is

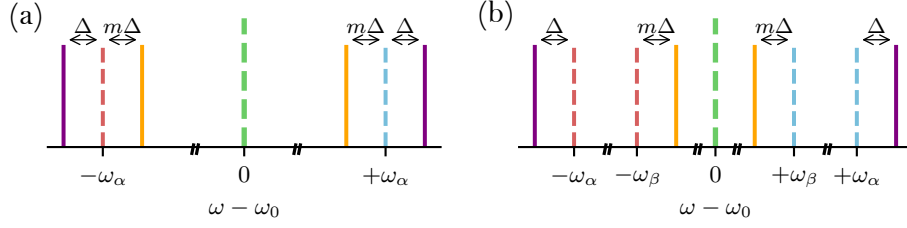


Figure 1.2: The quadchromatic field required for generating non-linear motional interactions via the two-SDF method. (a) For single-mode interactions, both SDF fields are detuned around the sidebands of the same motional mode, ω_α . Integer m is varied to select the desired interaction. (b) For two-mode interactions, each SDF is detuned from a different motional mode.

intractable for multiple-qumodes due to the large state-space, even with reasonably truncated oscillator representations (e.g. imposing a Fock-state cut-off). In the case of characterising systems consisting of multiple qubits and qumodes, either reduced tomography [41], or simplified metrics of performance [42] must be used.

1.1.2 Experimental Demonstration

In our group, we have previously demonstrated higher order squeezing [1], superpositions of non-linear bosonic states [2], beam-splitter [33], and two-mode squeezing interactions. However, due to the constraints of the previous experimental apparatus we were limited to global addressing fields, preventing the application of these hybrid interactions to more than two qubits. Here we demonstrate the creation and characterisation of a squeezed state on the newly constructed apparatus, described in chapter ??, utilising the single-qubit addressing system. Single addressing both increases the power efficiency of the interactions, and allows the control to select which qubit is used to couple to the motion — this provides a scalable route to generating hybrid interactions on larger qubit, and qumode registers.

Two Spin Dependent Forces

The individual SDFs are generated by the same bichromatic field method described in § ?? — i.e. simultaneous off-resonant blue- (BSB) and red-sidebands (RSB) of

motional mode ω_α are applied. To generate two simultaneously SDFs, we then require four optical fields with, frequency, phase, and amplitude control. The four required frequencies of the “quadchromatic” field are shown schematically in figure 1.2 with respect to the carrier frequency, ω_0 . Here, purple lines correspond to SDF_α , and orange lines to SDF_β . In this example we show the SDFs acting on a shared motional mode, but in general they may act on different modes. Similar to the generation of the bichromatic field, we may produce this quadchromatic field through the application of four RF tones onto a single-pass (SP) AOM. These tones are generated using an arbitrary waveform generator (AWG)¹. We cannot control the absolute optical phase at the ion, however we can set relative phases between the four fields by applying phase offsets between the RF tones with respect to some fixed time point. As discussed in § ??, the spin phase of the SDFs is set by the absolute phase of the RSB and BSB, $\hat{\sigma}_\phi = \cos(\phi_s)\hat{\sigma}_x + \sin(\phi_s)\hat{\sigma}_y$. To ensure we saturate the non-commutativity of the spin conditioning of the two SDFs, we require $\pi/2$ phase difference between the two spin phases,

$$\phi_{\alpha,s} - \phi_{\beta,s} = \frac{\phi_{\alpha,\text{RSB}} + \phi_{\alpha,\text{BSB}}}{2} - \frac{\phi_{\beta,\text{RSB}} + \phi_{\beta,\text{BSB}}}{2} = \frac{\pi}{2}. \quad (1.13)$$

In the following demonstration we set $\phi_{\alpha,\text{RSB}} = \phi_{\alpha,\text{BSB}} = \pi/2$, $\phi_{\beta,\text{RSB}} = \phi_{\beta,\text{BSB}} = 0$, however we note that for composing multiple hybrid operations into a circuit, the resulting motional phase of the SDFs can lead to further constraints on the chosen RSB and BSB phases.

The intensity of the quadchromatic field is ramped on and off with a rise envelope of $\sin^2(\pi t/(2t_{\text{ramp}}))$ applied via the AWG, where t_{ramp} is the ramp duration. This ramp suppresses excess off-resonant excitations of the A_1 phase loop term given in the previous section (further details in appendix A.8), and so we must select ramp duration based on the characteristic scale, $t_{\text{ramp}} \gtrsim \frac{1}{\Delta}$.

We aim to balance the power in each of the 4 tones. This may be calibrated using an optical pick-off before addressing the ion, directed onto a fast photodiode, and measuring the beatnote contrast when applying any pair of tones. In the following

¹*Sinara Phaser* board

squeezing demonstration, we balance the tones comprising the individual SDFs, but do not explicitly balance the power between SDFs, as the desired interaction scales as $\Omega_\alpha\Omega_\beta$, any imbalance leads to non-optimal interaction efficiency, but unlike imbalance of the RSB and BSB, does not increase coherent error terms. We observe that the duty cycle of the SP AOM can dramatically shift the power imbalance between all four tones, and so we take care ensuring that the total RMS power to drive one-, two-, and four-tones is the same as for the idle (no optical pulse) setting of the SP AOM, thus the only transient power drop is that of the ramp sequence.

Squeezing

We demonstrate the effective squeezing interaction on a single ion and radial motional mode. We are limited to using the radial modes due to the geometry of the single addressing system with respect to the trap. We use the upper radial mode with frequency, $\omega_{upper} = 2\pi \times 2.87$ MHz, and motional coherence $\tau_{upper} = 19(1)$ given in § ???. The initial near-vacuum motional state is prepared by Doppler and sideband cooling, with a typical thermal fock state distribution with mean $\bar{n} = 0.03(1)$, as described in § ??. The spin state is initialised into the ground state by optical pumping, described in § ??. By setting the relative detuning between the SDFs to $m = -1$, and ensuring anti-commuting spin conditioning by setting the relative phases of the quadchromatic field according to equation 1.13, we realise a $\hat{\sigma}_z$ -conditioned squeezing interaction, given by equation 1.8. As the spin-state is prepared in an eigenstate of the spin-conditioned squeezing, we expect squeezing on the qumode, without any change to the qubit state. We use a total quadchromatic beam power of 53 μ W, and set the base detuning, $\Delta = 2\pi \times 50$ kHz. We suppress excess excitation of the off-resonant error terms by using a ramp of $t_{ramp} = 40$ μ s. The state is characterised by the tomography method given in § 1.1.1, with a discretisation set to collect a 35×35 grid of displacement values, β , and $N = 300$ repeat samples to reduce shot noise. The full dataset for one of the tomographic scans takes around 1 hour to collect, and so we interleave mode calibration experiments

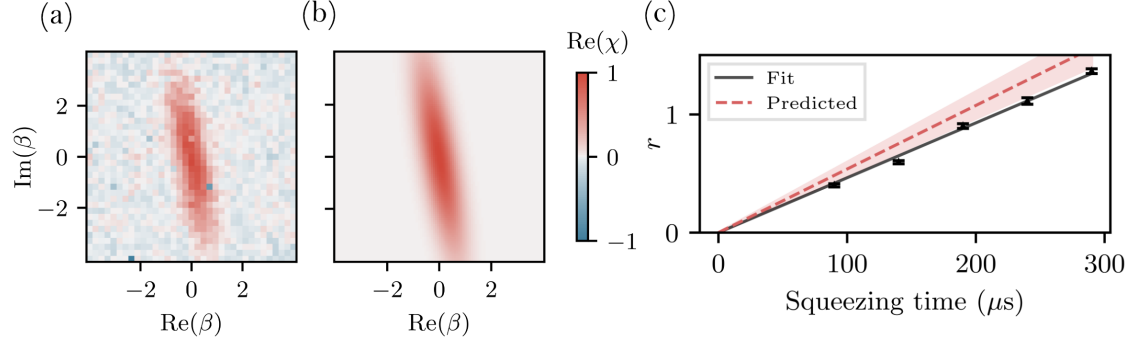


Figure 1.3: Demonstration of the squeezing interaction through the two-SDF method. (a) The characteristic function, $\text{Re}(\chi(\beta))$, of the squeezed state measured using the state tomography method given by figure 1.1. (b) The simulated state taking fit parameters inferred by equation 1.12. (c) The fitted squeezing parameter, r , as a function of two-SDF pulse duration.

(§ ??) to run every 5 minutes.

The on-resonance splitting SDF used for tomography must have its strength, Ω_{SDF} , calibrated to infer the magnitude of β applied. We use a total bichromatic beam power of 40 μW , and through fitting the $\hat{\sigma}_z$ expectation for the time-dependent splitting of a vacuum state,

$$\langle \hat{\sigma}_z \rangle = e^{-2|\beta|^2}, \quad |\beta| = \frac{\Omega_{\text{split}} t}{2}, \quad (1.14)$$

we measure an SDF splitting strength of $\Omega_{\text{split}} = 2\pi \times 7.2(4)$ kHz.

With these settings, the characteristic function for a squeezed state is measured, and the resulting real part is shown in figure 1.3 (a). We can see the distinctive contraction of one axis, and expansion of the other, as well as some small phase-space tilt, likely due to imprecise phase tracking between the squeezing interaction and the displacement probe. Fitting the data to equation 1.12 we may extract both the squeezing axis, ϕ_s , and the strength, r , where r is given by,

$$r = |\Omega_{\text{eff},S}|t = \frac{\Omega_\alpha \Omega_\beta}{\Delta} t. \quad (1.15)$$

An example fit is shown in panel (b), with extracted $r = 0.90(2)$, for a quadchromatic FWHM pulse duration of $t_{\text{sqz}} = 190$ μs . Panel (c) displays the squeezing strength as a function of t_{sqz} , with fit values given by the points, and the black line is a linear fit.

Assuming that the two SDF strengths are well balanced, $\Omega_\alpha = \Omega_\beta$, and taking the 53 μW used for the quadchromatic field compared to the SDF strength calibration at 40 μW , we find an expected squeezing strength of $\Omega_{\text{eff,S}} = 2\pi \times 0.86(9)$ kHz. This is shown as the red dashed line and shaded region. The fitted strength of $\Omega_{\text{eff,S}} = 2\pi \times 0.74(1)$ kHz is lower than the expected, suggesting that either we are not sufficiently balancing the strengths of the two simultaneous SDFs, or that the power stabilisation routine for 4-tones is inaccurate. Further exploration is needed, including careful quantification of the individual SDF strengths constituting the quadchromatic field. However, this preliminary result confirms that the motional coherence, and stability of the single-addressing system is sufficient to create, and characterise non-classical motional states using the apparatus developed for this thesis.

1.1.3 Outlook

Hybrid systems offer access to large Hilbert spaces, even with modest ion numbers. In this section we outlined a convenient set of native interactions for universal hybrid qubit-qumode gates. We experimentally demonstrate the squeezing interaction generated via the “two-SDF” method of reference [20]. This demonstration replicated previous work using the “two-SDF” method [1], however now shown on the newly built apparatus utilising a single-ion addressing system, and applied to the radial motional modes.

Enabled by the single-addressing, and improved motional coherence times of the new system, we are particularly interested in exploring short hybrid circuits composed of multiple interactions acting on an array of qubits and qumodes. Specifically, we are interested in utilising the hybrid system with increased robustness against errors through the use of quantum error mitigation methods applicable to both spin and oscillator subsystems. We outline one such method in the next section, with a preliminary demonstration on the spin subsystem.

Limitations

Using a single multi-ion chain to access multiple qubits and qumodes for scaling hybrid QC on the trapped-ion platform will be ultimately limited by spectral crowding of the motional modes, i.e. the average frequency spacing between neighbouring modes decreases as the number of accessible modes increases [43, 44]. The “two-SDF” method requires frequency selectivity to drive bosonic interactions exclusively on the chosen mode. To selectively drive a single mode, the magnitude of the detuning Δ must follow $\Delta \ll \delta\omega$, where $\delta\omega$ is the frequency difference to the nearest neighbouring mode. As discussed in the “two-SDF” method section, the strength of each SDF must be less than the base detuning, $\Omega_\alpha \ll \Delta$, to well approximate the effective non-linear unitary dynamics. As a result of these two requirements, the mode-frequency splitting directly limits the attainable strength of the non-linear interaction in the adiabatic regime, resulting in increasing gate duration with increasing register size.

A possible mitigation would be to rather scale the hybrid system using a collection of fixed-length chains with adequate mode-frequency splitting, and to transport quantum information between these chains either through ion transport whilst maintaining mode and spin coherence [45], generating remote entanglement through photonic interconnects [46], or through inducing collective motional interactions between separate ion-chains [47].

Another mitigation strategy could come from the pulse-shaping approach demonstrated for use in fast two-qubit entangling gates in the non-adiabatic regime [48], i.e. when $\Delta > \delta\omega$. Introducing extra degrees of freedom in the pulse amplitude envelope could allow for driving the desired interaction on the selected modes, while returning all spectator modes to their initial state by the end of the pulse. However, we note that the accumulation of non-global geometric phases, and possible non-commutativity of interactions on shared qubit-oscillator states could prohibit the use of such a pulse-shaping scheme. Ultimately, the desired interaction, mode frequency spectra, and target gate duration will determine the dominant coherent

errors. We leave such an analysis for future work.

1.2 Virtual Distillation

With access to noisy, intermediate-scale quantum information processors, the natural question is whether useful computations may be executed, without requiring full fault-tolerant error correction codes. In this section, we explore a resource-efficient error mitigation technique, known as virtual distillation, that can dramatically suppress the effect of errors on the final measured outcome for expectation value problems.

This section starts with outlining the goals of quantum error mitigation (QEM), with the following section giving details on the virtual distillation method. We then present an experimental demonstration suppressing errors on two-copies of a single qubit system. We finish this section with a discussion on future plans, and the extension of virtual distillation to hybrid CV-DV systems.

1.2.1 Error Mitigation

Noise coupled to a quantum system results in unintended coherent, and incoherent evolution. Deviations of the realised state from the target state are known as errors, and, if left unchecked, lead to the failure of quantum algorithms. This failure can either be due to a reduction of signal to noise ratio, or a bias in the measured expectation value, resulting in misleading conclusions. Ultimately, the goal of quantum error correction (QEC) is to detect and correct these errors through embedding information in larger Hilbert spaces [49]. Full QEC will allow for the execution of arbitrarily long algorithms ensuring errors (of up to a certain distance) do not accumulate over time. However, current QEC methods require large overheads in both qubit number and circuit depth, and are therefore technically challenging on current quantum hardware.

Furthermore, QEC is typically discussed with respect to discrete “gate-based” quantum computers, which although promise to be universal, pay for that privilege in often inefficient decompositions of target unitaries into a discrete set of gates. This poses an opportunity for near term devices to show quantum advantage using either short depth, uncorrected circuits [50], or leveraging CV systems to access larger computational spaces [10].

However, the impact of errors on uncorrected computation must be carefully managed. We consider algorithms that prepare some final state of interest and then take expectation-value measurements of some observable on that state. Uncertainty of the approximated expectation value due to averaging over a finite number of discrete samples, known as shot-noise, may be arbitrarily reduced by increasing the samples, N . However, due to imperfect execution of a given circuit, and coupling to the environment, there will be some systematic shift of the estimated expectation value compared to the ideal value of the noiseless state, ρ_{id} . This shift, or bias, of the approximated observable may be reduced by the application of quantum error mitigation (QEM) techniques [51, 52].

QEM protocols are varied in nature, making it difficult to give an overarching mechanism of operation. Reference [52] defines QEM as “... algorithmic schemes that reduce the noise-induced bias in the expectation value by postprocessing outputs from an ensemble of circuit runs, using circuits at the same noise level as the original unmitigated circuit or above.” This highlights the main distinction to QEC protocols: the physical noisy state is never corrected (often it is in fact degraded through the use of QEM) — rather the output from the noisy circuits is manipulated and postprocessed in such a way that we *effectively* recover an expectation value with suppressed deviation from the ideal value.

As a short example, consider a noisy state $\rho = \lambda\rho_{\text{id}} + (1 - \lambda)\rho_{\text{err}}$, where we have some probability of preparing the ideal state, and some probability of preparing an error state, ρ_{err} . We take N measurements of some observable on this mixed state, collecting n sample, $n = (n_0, n_1, \dots, n_N)$. Now consider the (unphysical) perfect QEM, consisting of an oracle that “knows” if each sample originated from

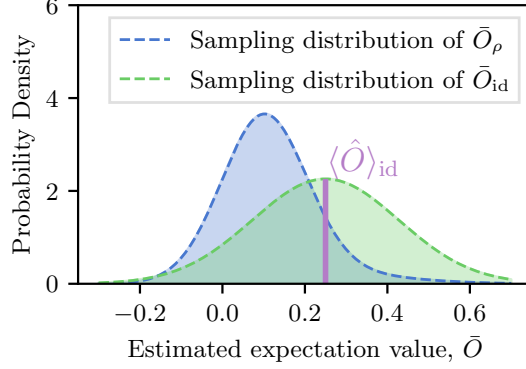


Figure 1.4: A schematic explanation of the goal and cost of quantum error mitigation. Deviations of the measured expectation value of an operator $\hat{O}$ on a noisy state, ρ , and on the ideal state ρ_{id} may be separated into components originating from shot noise, and from an underlying bias. Shot noise sets the variance of the distribution of estimated expectation values, while the underlying bias shifts the center of the distribution. Error mitigation reduces bias at the cost of increasing variance. Here we demonstrate this by a (non-physical) oracle, filtering out all samples originating from measurements of the non-ideal state to reconstruct a bias-free estimate of the ideal expectation value.

a measurement of ρ_{id} or of ρ_{err} , and flags them as such. In a postprocessing step, we may discard all samples associated with the error state, and retain the subset of samples of the ideal state, $N_{\text{id}} \subset N$. With this subset we may calculate the bias-free expectation value, however at the cost of increased shot noise, as $N_{\text{id}} \leq N$, and therefore increased variance of the estimated observable under many such experimental trials. This example, although fictitious, retains two important aspects of many QEM methods. First, we effectively “filter out” undesired information by some form of postprocessing, and second, this comes at a sample cost, where we effectively exchange a suppression of bias to an increase in shot noise. This trade-off is shown schematically in figure 1.4, for this oracle QEM.

Error mitigation protocols are diverse in their resource efficiency, assumptions on error-types, and scaling with system size. In this thesis, we consider the virtual distillation protocol [6–8] — a QEM method, applicable to both DV and CV systems, with few assumptions on the underlying noise model [53, 54]. It requires a fixed depth circuit appended to the probed state-preparation circuit, and a reasonable

resource overhead of $k \times$ more physical qubits (or qumodes) for a k -exponential decrease in error bias. With the apparatus constructed for this thesis, we may begin to explore this exponential error suppression for small system sizes. The following section gives details on the mechanism and practical circuits.

1.2.2 Virtual Distillation Protocol

Virtual distillation is a purification-based QEM, utilising k -copies of a circuit to reduce the final bias of the measured expectation value of a noisy state, ρ . The term *purification* is used to signify that the method effectively approximates the pure state closest to ρ , with the underlying assumption that the dominant eigenvector of ρ approximates the ideal pure state, $\rho_{\text{id}} = |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}|$. The validity of this approximation is explored in detail in reference [53], where they derive a bound for the deviation between the ideal state and dominant eigenvector, termed the “coherent mismatch”, c . This error provides the noise-floor for purification based techniques. Favourably, for many error models and circuit types, c is negligible to relevant applications, and in fact scales inversely proportionally to the number of qubits in a single copy, N , for fixed depth circuits.

The purified state may be accessed by,

$$\rho_{\text{pur}}(k) = \frac{\rho^k}{\text{Tr}[\rho^k]}, \quad (1.16)$$

where the denominator acts as a normalisation factor, and for $k = 2$ is the usual definition for the purity of a mixed state. To elucidate the claim that this is the form of the purified state, we expand the spectrum of ρ , assuming that the dominant eigenvector is the ideal state,

$$\rho = \lambda_{\text{id}} |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i |\psi_i\rangle \langle \psi_i|, \quad (1.17)$$

with $\lambda_{\text{id}} > \lambda_i$ for all i , and from normalisation, $\lambda_{\text{id}} + \sum_{i=1}^{2^N-1} \lambda_i = 1$. Inserting this expansion into equation 1.16, we find,

$$\rho_{\text{pur}}(k) = \frac{1}{\mathcal{N}} \left(\lambda_{\text{id}}^k |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}| + \sum_{i=1}^{2^N-1} \lambda_i^k |\psi_i\rangle \langle \psi_i| \right), \quad (1.18)$$

with normalisation factor $\mathcal{N} = \lambda_{\text{id}}^k + \sum_{i=1}^{2^N-1} \lambda_i^k$. The exponentiation acts to relatively attenuate all error eigenvalues compared to λ_{id} , with the limit in k recovering the pure ideal state, $\lim_{k \rightarrow \infty} \rho_{\text{pur}}(k) = |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}|$.

The remaining question is how can we experimentally access $\rho_{\text{pur}}(k)$? In references [7, 8], it is shown that the expectation value of the purified state with respect to the applied operator $\hat{O}$ is,

$$\langle \hat{O} \rangle_{\text{pur}} = \text{Tr}[\hat{O} \rho_{\text{pur}}(k)] = \frac{\text{Tr}[\hat{O} \rho^k]}{\text{Tr}[\rho^k]}, \quad (1.19)$$

and crucially may be estimated without ever physically preparing $\rho_{\text{pur}}(k)$. Instead the numerator and denominator may be found through preparing k -copies of the noisy state, applying some derangement operation to entangle all copies together, and measuring the desired observable on one of the copies. This can be understood through the following identity² providing the numerator,

$$\text{Tr}[\hat{O} \rho^k] = \text{Tr}[\hat{O}^{(i)} \hat{S}^{(k)} \rho^{\otimes k}], \quad (1.20)$$

where $\hat{S}^{(k)}$ is the derangement operator, which permutes the k copies of the state, and $\hat{O}^{(i)}$ is the operator acting on the i th copy. The denominator of the corrected expectation value is found by replacing the operator with the identity operator. There is some freedom in the form of the derangement, here we shall give it explicitly as the cyclic permutation of k objects,

$$\hat{S}^{(k)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_k\rangle = |\psi_2\rangle \otimes |\psi_3\rangle \otimes \dots \otimes |\psi_k\rangle \otimes |\psi_1\rangle. \quad (1.21)$$

Practically, if we can perform this multi-copy derangement, $\hat{S}^{(k)}$, and single-copy operator $\hat{O}^{(i)}$, on k -copies of the noisy state, we can postprocess a collection of circuit repetitions to recover the k th purified expectation value $\langle \hat{O} \rangle_{\text{pur}}$.

In the following section we provide two possible approaches for implementing these unitaries and measurements, and give explicit circuit decompositions for both. In § 1.2.3 we explore the scheme experimentally, and particularly study the breakdown of error suppression due to correlated noise *between* copies. We give

²Huggins [8] provides a proof of the identity, which we give an example for the case $k = 2$ in appendix A.9.

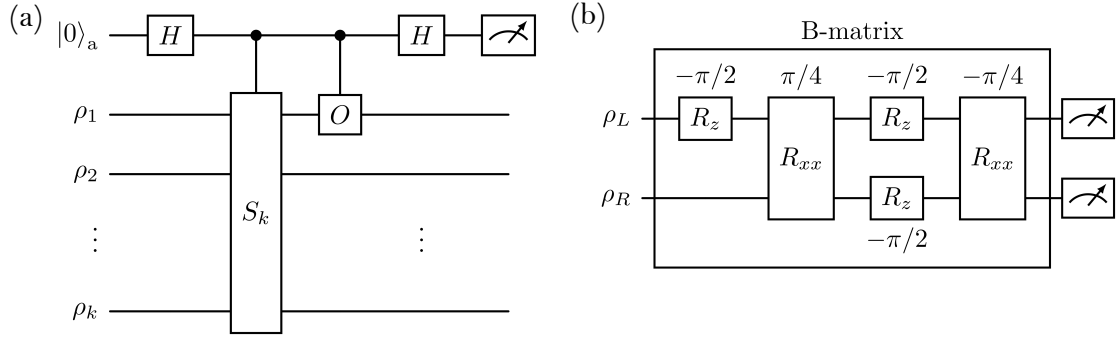


Figure 1.5: Two circuits for implementing virtual distillation. (a) Hadamard test method using controlled derangement, $\hat{S}^{(k)}$, over k copies, and controlled operator, $\hat{O}$, applied on one copy. The ancilla qubit is measured to extract $\text{Tr}(\hat{O}\rho^k)$. Repeating the circuit without the controlled operator measures normalisation factor, $\text{Tr}(\rho^k)$, used to recover the virtually purified result, $\langle \hat{O} \rangle_{\text{pur}}$. (b) Two-copy implementation using a B -circuit, here decomposed into two entangling gates and single qubit phase gates. Measurement of both qubits in the computational basis, and classical postprocessing recovers $\text{Tr}(\hat{Z}\rho^2)$ and purity, $\text{Tr}(\rho^2)$, used to recover $\langle \hat{Z} \rangle_{\text{pur}}$.

details of the impact on suppression when the underlying independence assumption of the errors on each noisy copy is violated.

Circuits

We discuss two methods of implementing virtual distillation. The first being a general approach able to act on arbitrary Hilbert spaces using Hadamard tests, and the second being more resource efficient, however not simply generalisable to arbitrary sizes of each copy, and to multi-qubit observables.

We begin with the Hadamard test, to build further intuition on virtual distillation based on similar circuit topology discussed in § 1.1.1 for CV-tomography. The circuit for virtual distillation via an ancilla is shown in figure 1.5 (a). The derangement and desired operator, $\hat{O}$, are applied conditioned on the ancilla qubit, with a final measurement only made on the ancilla. The circuit is repeated for N_{shot} times and the outcomes are averaged to recover the estimated real part of the numerator in equation 1.19. The circuit is again repeated N_{shot} times with the controlled operator replaced with the identity operator. Subsequent averaging recovers the denominator, which may then be used to recover $\text{Re}[\langle \hat{O} \rangle_{\text{pur}}]$.

The controlled derangement operation may be decomposed into a series of controlled-swap gates [7], which for DV systems may be decomposed to native single and two-qubit gates accessible on the ion trap platform. For CV-systems, the derangement can be decomposed to a series of Fredkin gates [55, 56], which in turn may be accessible by either controlled-beam-splitters [21, 57] and/or controlled-parity operations [58]. These interactions are available in our system by the “two-SDF” method, as highlighted in table 1.1.

The second method, known as the “ B -matrix” [8], is slightly more resource efficient, not requiring ancilla qubits, however does not simply scale for $k > 2$, and for non- $\hat{Z}$ measurements. The approach recognises that for $k = 2$ the symmetrised computational observable $\hat{Z}_2 = \frac{1}{2}(\hat{Z}\hat{I} + \hat{I}\hat{Z})$ commutes with the SWAP operator, $\hat{S}^{(2)}$, and consequently so too do both observables required for virtual distillation,

$$[\hat{S}^{(2)}, \hat{Z}_2 \hat{S}^{(2)}] = 0. \quad (1.22)$$

Thus the two observables possess a common eigenbasis and may be simultaneously diagonalised by a single matrix³,

$$\hat{B} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}. \quad (1.23)$$

In the computational eigenbasis, the observable is now given by,

$$\langle \hat{A} \rangle = \text{Tr}[\hat{B}^\dagger \hat{D} \hat{B} \rho], \quad (1.24)$$

$$= \sum_{x \in \{0,1\}^2} d_x p(x), \quad (1.25)$$

where $\hat{A}$ is either $\hat{Z}_2$ or $\hat{S}^{(2)} \hat{Z}_2$, and the second line follows by the definition of the diagonalised operator $\hat{D} = \sum_x d_x |x\rangle \langle x|$, and $p(x) = \langle x | \hat{B} \rho \hat{B}^\dagger | x \rangle$ is the probability of observing computational state $|x\rangle$ after the basis change given by $\hat{B}$.

³Note, we use a different convention than reference [8]. This is done for a simpler decomposition to native gates on our apparatus.

As a result, both numerator and denominator from equation 1.19 may be simultaneously measured by computational basis measurements after applying a two-qubit basis rotation⁴. The values are related to computational expectation values by,

$$\text{Tr}[\hat{Z}\rho^2] = \frac{1}{2}(1 - \langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle + \langle \hat{Z} \otimes \hat{Z} \rangle), \quad (1.26)$$

$$\text{Tr}[\rho^2] = \frac{1}{2}(\langle \hat{Z} \otimes \hat{I} \rangle + \langle \hat{I} \otimes \hat{Z} \rangle). \quad (1.27)$$

Figure 1.5 (b) shows the B -matrix decomposition into native gates to our platform. A demonstration of this simple two-copy example of virtual distillation is given in the following section.

1.2.3 Experimental Demonstration

This section begins with experimental details on implementing virtual distillation via the B -matrix decomposed to native two-qubit R_{xx} gates and single qubit phase shifts, R_z . With this protocol we can then study the efficacy of virtual distillation in two different noise regimes — independent depolarisation, and correlated errors due to common random-unitaries (see § ?? for the noise model). This work provides evidence of the performance of virtual distillation under realistic structured errors originating from fluctuations of a common external field.

Virtual Distillation on Two Copies

We demonstrate virtual distillation via the B -circuit shown in figure 1.5 (b) on two-copies of a single qubit. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ of a $^{40}\text{Ca}^+$ ion in a quantising field of ~ 5 G. Gates are driven using 729-nm light focussed onto the ions by the individual addressing system detailed in § ??. Two-qubit entangling gates of the form R_{xx} are implemented using Walsh-1 MS gates on the in-phase upper radial mode, as described in § ??. For this decomposition, we specifically require $R_{xx}(\frac{\pi}{4})$ -gates. To calibrate these gates, two consecutive $R_{xx}(\frac{\pi}{4})$ pulses are applied, and a partial state-tomography protocol [59] is utilised (as summarised by equation ??).

⁴Although not discussed here, this method scales simply to $\hat{Z}$ measurements on copies containing multiple qubits through parallel pairwise rotations [8].

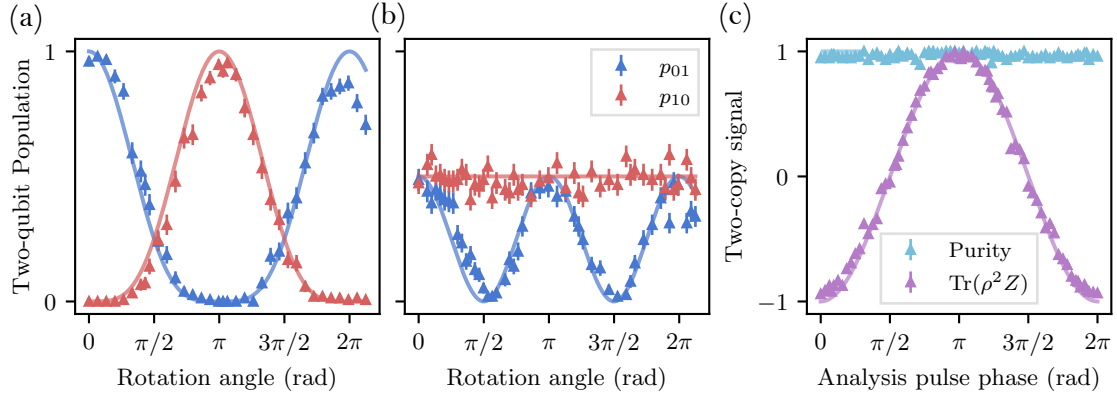


Figure 1.6: Verification of the physically implemented B -circuit. (a-b) Preparing the initial state $|01\rangle$, applying global $R_x(\theta)$ rotations, and measuring the two-qubit joint probabilities provides a clear signal for the correct application of the B -matrix. (a) No B -circuit is applied. (b) B -circuit applied. (c) Globally applied Ramsey experiment followed by the B -circuit and subsequent postprocessing to recover the purity and numerator term given by equations 1.26.

After calibration, the Walsh-1 $R_{xx}(\frac{\pi}{4})$ gate settings are the following: 21 μW total 729-nm beam power, -15 kHz detuning from in-phase upper radial motional mode, 10 μs ramp duration (ramped on and off for both arms of the Walsh), and a 67.5 μs FWHM pulse duration for each arm. The total pulse duration including ramp durations is then 155 μs . Stark-shifts must be carefully calibrated to ensure consecutive MS gates have spin-phases set relative to the qubit. The Stark-shift for this power is calibrated using an SDF on a single ion (described in § ??), finding a shift of 2.49 kHz. This phase is accounted for by updating the RF-tones on the SP AOM for subsequent pulses. With these settings we measure a fidelity for two-consecutive $R_{xx}(\frac{\pi}{4})$ gates of $\mathcal{F}_2 = 96.0(8)\%$, with population component of $p_{01} + p_{10} = 0.010(3)$, and parity contrast of $C = 0.93(2)$. The inferred single $R_{xx}(\frac{\pi}{4})$ gate fidelity, assuming depolarising errors, is then $\mathcal{F} = 98.0(4)\%$.

Single qubit phase shifts, R_z , are executed virtually. As we require non-common R_z phase shifts on the two qubits, we update the relative phases of the two pairs of RF tones applied to the acousto-optic deflectors, used for generating the addressed spots. As these single-qubit gates are virtual, we expect negligible error contributions.

The final computational-basis measurements are executed by state-dependent fluorescence, described in § ??. From the single-shot outcomes, the joint probabilities

p_{ij} , with $i, j \in \{0, 1\}$ denoting the bright and dark states of ions 1 and 2 are recovered, respectively. The required expectation values given in equations 1.26 are then simply,

$$\langle \hat{Z} \otimes \hat{I} \rangle = p_{00} + p_{01} - p_{10} - p_{11}, \quad (1.28)$$

$$\langle \hat{I} \otimes \hat{Z} \rangle = p_{00} - p_{01} + p_{10} - p_{11}, \quad (1.29)$$

$$\langle \hat{Z} \otimes \hat{Z} \rangle = p_{00} - p_{01} - p_{10} + p_{11}. \quad (1.30)$$

To initially verify the performance of the physical implementation of the B -circuit, two heuristic tests are used⁵. First, we want a clear indication that the transformation due to the applied B -circuit matches the convention given by the B -matrix in equation 1.23. Inspecting the B -matrix, we see the central block acts distinctly on the $\{|01\rangle, |10\rangle\}$ subspace. As such, a method that preferentially probes this subspace would give a clear signal for conformity between the applied B -circuit and the desired B -matrix. To access this subspace, a singly addressed $R_x(\pi)$ -pulse is applied to prepare the initial state $|01\rangle$. A global $R_x(\theta)$ is then applied with variable rotation angle, θ . Figure 1.6 (a) shows the measured two-qubit populations as a function of θ without the B -circuit applied. Panel (b) shows the populations with the global rotation and the B -circuit applied. The solid lines correspond to the expected populations inferred from equation 1.23, while points are experimental data with $N = 150$ samples per probed rotation angle. We can see good qualitative agreement between the observed data, and the desired behaviour suggesting the phase relationships enforced by the B -circuit indeed conform with the ideal B -matrix.

The second heuristic check verifies the conventions for recovering the numerator and denominator in equations 1.26, from the qubits measured joint probabilities, given in equations 1.28. To test this, a simple Ramsey experiment is performed on both qubits simultaneously. The qubits are initially prepared in $|00\rangle$. A global pulse $R_x(\frac{\pi}{2})$ is then applied, followed by a final analysis pulse $R_\phi(\frac{\pi}{2})$, applied with variable spin-phase, ϕ . After these pulses, the B -circuit is applied, and the two-qubit

⁵Process-tomography would be ideal for this two-qubit implementation, however we have not yet implemented this on our apparatus.

populations are measured. As the two qubits remain in identical pure-states to one another (up to errors in the parallel single-qubit gates), we expect the inferred purity to remain unity, $\text{Tr}[\rho^2] = 1$. Given this, we expect the recovered numerator term to be equivalent to typical Ramsey oscillations on a single qubit,

$$\text{Tr}[\hat{Z}\rho^2] = \langle \hat{Z} \rangle_\phi = -\cos(\phi). \quad (1.31)$$

Figure 1.6 (c) displays the post-processed purity and numerator terms extracted as a function of analysis pulse phase, ϕ , with solid lines showing the expected behaviour of the ideal B -circuit. Again we see excellent conformity between experimental and desired behaviour, giving evidence that we are successfully applying virtual-distillation.

In the next section we use virtual distillation for its intended purpose — to suppress the impact of errors. We will discuss the efficacy of virtual distillation under two different noise models, providing experimental evidence for both.

Virtual Distillation under Correlated Errors

To quantify the achieved suppression of errors through the use of virtual distillation, we must characterise and inject errors with a known structure. As such, applying randomised Clifford sequences, the same as those used in standard randomised benchmarking (RB), are a convenient method for known error injection. We frame the task of quantifying error suppression in the context of single-qubit RB by comparing the exponential decay due to some error-per-Clifford, ε_{raw} , to the effective decay profile of the error-mitigated observable, $\langle \hat{Z} \rangle_{\text{pur}}$.

Figure 1.7 (a) shows the circuit implemented for controlled error injection — random Clifford sequences with a final inverting gate are applied in parallel to both qubit copies, effectively driving depolarising channels with total circuit error rate variable by sequence length, m . The B -circuit is then applied followed by $\hat{Z}$ measurements. Postprocessing of the two-qubit joint probabilities recover the error-suppressed expectation value, $\langle \hat{Z} \rangle_{\text{pur}}$.

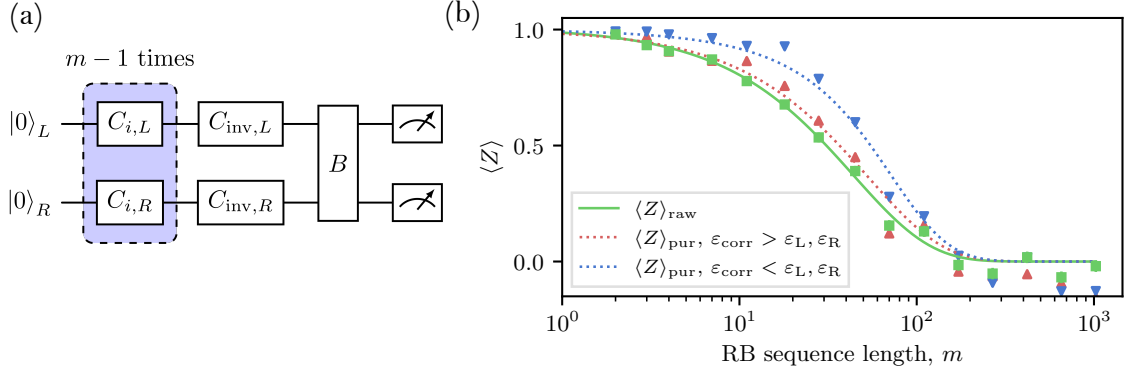


Figure 1.7: Verifying the performance of virtual distillation under either dominant local depolarising or correlated errors. Panel (a) shows the circuit of injected errors followed by the B -circuit. Panel (b) shows the effective suppression of errors in the single-qubit $\hat{Z}$ observable using virtual distillation on two-copies. The green points correspond to the raw single-qubit decay due to the physical errors, with the solid green line being an exponential fit. The dotted lines correspond to the expected error mitigated single qubit decays inferred by the model given in equation 1.34, and the error-per-Cliffords measured by k -copy RB. The red and blue points correspond to the error mitigated $\hat{Z}$ observables for in the regime of dominant correlated errors and local depolarising errors respectively.

The RB protocol is repeated without the applied B -circuit to recover the standard single-qubit exponential decays, $\langle \hat{Z} \rangle_{\text{raw}}$, and the corresponding error-per-Clifford, ε_{raw} .

Crucially, with this circuit we may explore two different error regimes: first, we may apply independent single-qubit Clifford sequences to each copy, which independently twirls the error channel acting on each qubit. This decorrelates errors between the two copies [60, 61] leading to favourable performance of virtual distillation.

In contrast, with the second regime we purposefully apply the same Clifford sequence to each qubit, i.e. by enforcing $C_{i,L} = C_{i,R}$. In this regime, any common-mode coherent errors acting on both qubits will not be decorrelated by the parallel twirling (see k -copy twirling in chapter ??). These correlated errors can significantly degrade the performance of virtual distillation. We shall spend some time now to make this connection concrete.

Consider the k -copy mixed state, $\rho^{(k)}$, for a common-mode error channel,

$$\rho^{(k)} = \sum_i \kappa_i \rho_i^{\otimes k}. \quad (1.32)$$

Applying virtual distillation, using equation 1.19, and the identity equation 1.20, the effective measured observable is,

$$\langle \hat{O} \rangle_{\text{pur,c}} = \frac{\sum_i \kappa_i \text{Tr}[\hat{O} \rho_i^k]}{\sum_i \kappa_i \text{Tr}[\rho_i^k]}, \quad (1.33)$$

where crucially the trace operator is now within the sum. Clearly, the components κ are not exponentiated in the same manner as the λ coefficients of the previously purified density matrix, ρ_{pur} , in equation 1.18 — as such the dominant state is no longer effectively amplified through derangement. In the specific case of common random unitary noise, discussed in § ??, ρ_i remains a pure state, $\rho_i = U_i |\psi_{\text{id}}\rangle \langle \psi_{\text{id}}| U_i^\dagger$, as such the effective observable reduces to $\langle \hat{O} \rangle_{\text{pur,c}} = \text{Tr}[\hat{O} \rho] = \langle \hat{O} \rangle_{\text{raw}}$, where $\langle \hat{O} \rangle_{\text{raw}}$ is the expectation value of a single copy, $\rho = \sum_i \kappa \rho_i$. For this particular correlated error channel, virtual distillation is entirely ineffective.

Such correlated errors are commonplace in the ion-trap platform due to noisy global control fields interacting with the ions. Under parallel Clifford twirling, these global control field errors are simplified such that they can be quantified by a single value “correlated error-per-Clifford”, $\varepsilon_{\text{corr}}$, as discussed extensively in § ??. As such, the relative proportions of uncorrelated, depolarising type errors, to correlated *common noise* errors is required for predicting the performance of virtual distillation. The task of quantifying these errors is the subject of chapter ??, and we shall use the developed *k-copy* RB protocol here.

The analytic decay of the mitigated observable $\langle \hat{Z} \rangle_{\text{pur}}$ when the two-copies are subject to a shared error channel, Λ_T , consisting of *k-copy* twirled common random unitaries (parameterised by $\varepsilon_{\text{corr}}$) and local depolarising errors on the two copies (parameterised by ε_L and ε_R) is given by,

$$\langle \hat{Z} \rangle_{\text{pur}} = (1 - 2\varepsilon_{\text{corr}})^m \frac{\cosh(m(\varepsilon_L - \varepsilon_R))}{\cosh(m(\varepsilon_L + \varepsilon_R))}, \quad (1.34)$$

where we have assumed small errors. This is simply calculated by extending equation ?? to include the action of the B -matrix, here using the superoperator form,

$$p_{LR} = \langle \langle L, R | B \Lambda_T^m | 00 \rangle \rangle, \quad (1.35)$$

Twirling type	$\varepsilon_L/10^{-3}$	$\varepsilon_R/10^{-3}$	$\varepsilon_{\text{corr}}/10^{-3}$	$\varepsilon_{\text{raw}}/10^{-3}$
Independent	7(1)	7(1)	4(1)	11(2)
Identical	2.0(4)	1.4(4)	9.3(5)	11.2(6)

Table 1.2: Local depolarising and correlated errors contributions of both the independently and identically twirled channels on two copies of a single-qubit. These error-per-Cliffords are characterised by the *k-copy* RB method described in § ??.

where p_{LR} is the probability of measuring bit string $LR \in \{0, 1\}^2$. The effective observable $\langle \hat{Z} \rangle$ is then recovered by equations 1.28. For small m , and assuming small errors, this may be expanded to second order as,

$$\langle \hat{Z} \rangle_{\text{pur}} \approx 1 - 2m\varepsilon_{\text{corr}} - 2m^2\varepsilon_L\varepsilon_R, \quad (1.36)$$

where we can see that correlated errors maintain a linear decay, while depolarising errors are suppressed to a quadratic decay. We can also see that in the ideal case of exclusively depolarising type errors of equal magnitude, the effective observable decays as $\langle \hat{Z} \rangle_{\text{pur}} = \text{sech}(2m\varepsilon_L)$.

We demonstrate this behaviour experimentally by applying interleaved single-qubit Clifford gate sequences on two qubits followed by the derangement protocol given in the previous section. The interleaved sequences are implemented by alternating the position of the single-ion addressing beam after each Clifford, requiring a delay of $\approx 10 \mu\text{s}$ between gates. To ensure a baseline correlated unitary error is present, the single-qubit gates comprising the Clifford sequence are driven off-resonantly by $\delta = 2\pi \times 1 \text{ kHz}$. For both identical and independent Clifford twirling, the correlated and depolarising errors are quantified by *k-copy* RB, with errors summarised in table 1.3. The different twirling approaches change the distribution of errors between correlating and depolarising, while retaining a consistent total single-qubit error rate. It is interesting to note that the independent twirling is not sufficient in this case to entirely decorrelate the errors between the two qubits. These errors are used to predict the effective error-mitigated decay profiles, and are plotted as a function of sequence length, m , in figure 1.7 (b). The RB sequences

Copy Num, k	Total $R_{xx}(\frac{\pi}{4})$	Total $R_x(\frac{\pi}{2})$	Expected Faults
$2(B)$	2	0	0.02
2	22	7	0.117
3	33	14	0.234
4	44	21	0.351
5	55	28	0.468

Table 1.3: Required number of $R_{xx}(\frac{\pi}{4})$ and $R_x(\frac{\pi}{2})$ gates to implement the derangement circuit for virtual distillation on multiple copies. Apart from $n = 2(B)$, which uses the B -circuit, all other rows are using the ancilla-based Hadamard test circuit, implemented by controlled-SWAP gates. We give the expected total faults assuming fidelities of 99% for two-qubit operations, and 99.9% for single qubit operations.

are repeated with the B -circuit applied, and the postprocessed $\langle \hat{Z} \rangle_{\text{pur}}$ are plotted. We can see that there is good agreement between the expected mitigated decay profiles and the observed data. This demonstration highlights the negative impact of correlated errors on the performance of virtual distillation, as well as the importance of sufficiently decorrelating errors, which we do here by local Clifford twirling.

1.2.4 Outlook

Quantum error mitigation is accessible to near-term quantum computers, under some circumstances allowing for resource-efficient, and substantial reductions in measurement bias. In this section we have provided some background on quantum error mitigation via virtual distillation, highlighting two possible implementations: a Hadamard test, and the B -circuit. We demonstrate the B -circuit on the trapped-ion platform, and use the protocol to explore the ability to suppress errors under unfavourable correlated error channels. These demonstrations provide the groundwork for two future research directions — exploring the practical achievable advantage when extending to more than two copies; and extending to error-mitigation acting on a hybrid system.

Extending to > 2 copies

From equation 1.18, we can see that increasing the copy number should exponentially approximate the dominant pure state. We want to observe this exponential

behaviour, and so want to scale past two copies. To do this, we will use the ancilla-based Hadamard circuit, as shown in figure 1.5 (a).

For n copies of noisy state, ρ , the controlled derangement can be decomposed to a minimum of $n - 1$ controlled-SWAP gates [62]. Using accessible native gates, these controlled-SWAP are decomposed to $n_{\text{MS}} = 11 R_{xx}(\frac{\pi}{4})$ gates and at most $n_{\text{SQ}} = 7$ single qubit $R_x(\frac{\pi}{2})$ rotations. Assuming that we can implement $R_{xx}(\frac{\pi}{4})$ with fidelity, $\mathcal{F} = 99\%$, and single qubit gates with fidelity, $\mathcal{F} = 99.9\%$, we may calculate the expected total faults per circuit for a given copy size. Here, we assume the error on each gate is independent, and define expected fault number as $N_f = n_i(1 - \mathcal{F}_i)$. Table 1.3 summarises the gate scaling and expected fault number for the derangement operation. The first row gives the two-copy method via the B -circuit for reference.

To effectively suppress errors, we (heuristically) desire that the derangement operation contributes less faults than the state-preparation of the noisy copy. Further, we desire that the total circuit has at most $N_f \approx 1$. From the table, we see that although it might be possible to demonstrate up to $k = 5$ with our expected gate errors, this might be currently out of reach due to the large error contribution from the derangement. The difficulty in predicting the efficacy of the noisy derangement is that the impact of the errors on the performance of virtual distillation highly depends on their structure. Practically observing the suppression of circuit errors with copy number k , will thus provide evidence of the impact of individual gate errors on the derangement. It may also be useful to further characterise the individual gate errors using tomographic methods to predict the performance of the noisy-derangement.

Virtual Distillation on Hybrid Systems

Hybrid systems, as discussed in § 1.1, offer advantages in computation, due to the large accessible state space, with low numbers of physical information carriers. Quantum harmonic oscillators (qumodes) also have applications in quantum simulation [63] and sensing [64, 65]. However, due to environmental and control

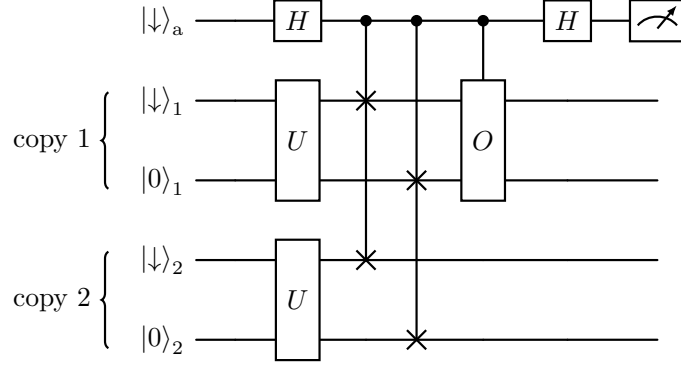


Figure 1.8: Proposed circuit for applying virtual distillation to a hybrid system. Using the ancilla method for measuring $\text{Tr}(\hat{O}\rho^k)$, we may decompose the controlled derangement to the transversal application of Fredkin gates acting on either qubit or qumode subsystems.

noise, the performance and reliability of these systems has strong limitations. Virtual distillation offers a resource efficient method for suppressing errors in CV systems if a pure state is targeted. The proposal for virtual distillation originated from measuring an observable on a thermal qumode state effectively prepared at a lower temperature than the constituent noisy copies [6].

We want to extend this protocol to a register of qubits and qumodes, applying transversal derangement, as seen in figure 1.8, to suppress errors in hybrid systems. Particularly, the mitigation of dephasing and Gaussian errors on the qumode is an under-explored topic, and poses an opportunity for greatly reducing the bias in quantum simulation tasks [66].

To access this, we require a Fredkin (controlled-SWAP) operation acting between an ancilla qubit and two qumodes. For certain Fock-state distributions, this may be realised by a single controlled-beamsplitter interaction [21], which we have simple access to via the “two-SDF” method in § 1.1.1. However, for the general case, we require both parity and beamsplitter operations, with control on either gate [67]. Demonstrating the Fredkin on a hybrid system is also of particular interest due to its reversible nature, and the practical advantage that it conserves the total

excitation number of the system [55].

Limitations

Error mitigation is resource efficient, but only allows access to measurements of an *effectively* error suppressed state, ρ_{pur} — entropy is never physically removed from the system, as is the case for QEC. The required samples for a fixed variance of the measured observable scales exponentially with the expected total number of faults in the circuit, N_f [52]. As such, QEM is only practically advantageous up to $N_f \approx 1$ (although this is somewhat a heuristic bound, and depends on many implementation details). QEM is thus suited for near-term applications on non-fault tolerant short depth circuits.

As shown in § 1.2.3, correlated and structured errors between copies can greatly reduce the efficacy of virtual distillation. This may limit the practicality of virtual distillation on some experimental platforms relying on global addressing fields. This may be somewhat mitigated either through appropriate independent twirling of the copies or through random circuit compositions [61]. Furthermore, physically separating the copies using networked modules can ensure that errors originating from external environmental factors are no longer common-mode between copies [62].

2

Conclusion

Appendices

A

Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Gate Fidelity

We derive the fidelity of a noisy gate implementation with respect to the ideal unitary.

A quantum channel,

$$\mathcal{E} : \rho \mapsto \rho, \tag{A.1}$$

which executes ideal unitary, U , with some error, has an average operation fidelity given by,

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \tag{A.2}$$

where the integral is taken over the uniform Haar measure on the space of pure states [68, 69]. We shall see that the language used for this analysis of single qubit

gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right), \quad (\text{A.3})$$

where V is the error unitary, and d is the dimension of the Hilbert space.

Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.2 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2, \end{aligned} \quad (\text{A.4})$$

where $W = U^\dagger V$. Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.5})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.6})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These

two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.7})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)} (\mathbb{I} + S). \quad (\text{A.8})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.9})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter ???. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.3.

A.6 Projectors of $Cl_1^{\otimes 2}$

We outline the method of recovering the α parameters described in section ??? using projectors onto the irrep spaces. A special thanks to Iver Øvergaard who performed this analysis applied to the *common noise* channel and gave permission for reproducing the results here **IRO please check you are happy with me to include this appendix.**

The decay due to a single rotation acting identically on both qubits which, in the Pauli-transfer matrix (PTM) representation, can be written as $\Lambda^{\mathcal{P}} = \text{diag}(1, R, R, R \otimes R)$, where superscript $\mathcal{P}$ represents the block diagonal matrix is ordered by the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, with $\vec{\sigma} = \{\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$, and

$R = R(\vec{n}, \theta) \in SO(3)$ describes a rotation by θ around $\vec{n}$. Finding the vector $\vec{\alpha}$ can now be done using Schur's lemma for the case of no mixing between irrep spaces:

$$\tilde{\Lambda}^{\mathcal{P}} = \sum_{\lambda} \alpha_{\lambda} \Pi_{\lambda}, \quad \alpha_{\lambda} = \frac{1}{d_{\lambda}} \text{Tr}(\Pi_{\lambda} \Lambda^{\mathcal{P}}). \quad (\text{A.10})$$

Here, Π_{λ} denotes the PTM projector onto the irrep space labelled λ and with dimension d_{λ} . It follows immediately that the identity sector is fixed, while the left- and right-qubit sectors produce $\alpha_L = \alpha_R = \text{Tr}(R)/3$. In principle, α_{λ} for the two-qubit sector can be evaluated by explicitly constructing the projectors from the basis operators in Table ???. However, in this case the symmetries of these irrep spaces allow the projectors to be expressed more simply in terms of operators whose traces are easily evaluated:

$$\Pi_{tr} = T, \quad (\text{A.11})$$

$$\Pi_{dt} = D - T, \quad (\text{A.12})$$

$$\Pi_{so} = \frac{1}{2}(I_9 + S) - D, \quad (\text{A.13})$$

$$\Pi_{as} = \frac{1}{2}(I_9 - S), \quad (\text{A.14})$$

where $T = 1/3 \sum_{ij} |ii\rangle \langle jj|$, $D = \sum_i |ii\rangle \langle ii|$, and $S = \sum_{ij} |ji\rangle \langle ij|$ are the trace projector, diagonal projector, and swap operator, respectively. The required traces are

$$\text{Tr}(I_3 R) = \text{Tr}(R) = (1 + 2c), \quad (\text{A.15})$$

$$\text{Tr}(I_9(R \otimes R)) = (\text{Tr}(R))^2 = (1 + 2c)^2, \quad (\text{A.16})$$

$$\text{Tr}(T(R \otimes R)) = \frac{1}{3} \text{Tr}(R^T R) = 1, \quad (\text{A.17})$$

$$\text{Tr}(D(R \otimes R)) = \sum_i R_{ii}^2 = c^2 + 2c + (1 - c)^2 \kappa, \quad (\text{A.18})$$

$$\text{Tr}(S(R \otimes R)) = \text{Tr}(R^2) = 4c^2 - 1, \quad (\text{A.19})$$

where $c = \cos(\theta)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Substituting these traces into equation A.10

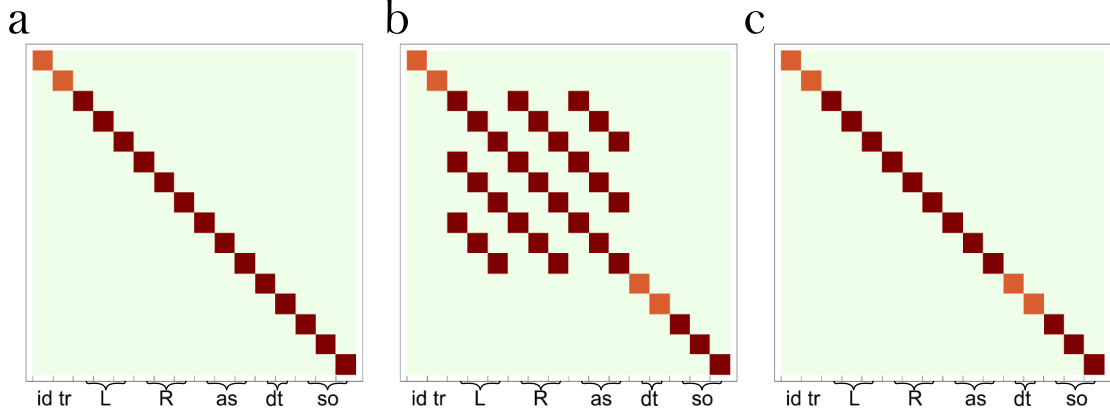


Figure A.1: Liouville operator representation of error channels after twirling over $Cl_1^{\otimes 2}$, ordered by the irrep basis (Schur basis). (a) *Common noise*, (b) *Continuous ZZ*, (c) *Stochastic ZZ*. Orange blocks represent an entry of 1, green represent 0, and red represent a decay with dependence on m .

and using equation A.11 yields

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (\text{A.20})$$

$$\alpha_{tr} = 1, \quad (\text{A.21})$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2\kappa), \quad (\text{A.22})$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2\kappa). \quad (\text{A.23})$$

A.7 ZZ-type noise

In the main text we explore a separable error channel that induces correlations between qubits in a register. Here we shall explore entangling type noise, which although not prevalent on the ion-trap qubit platform, is a dominant error source in superconducting qubits.

The always on ZZ-type error common to superconducting qubits is described by the Hamiltonian,

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z, \quad (\text{A.24})$$

where J_{ZZ} is the coupling strength. Generally, H_{ZZ} will couple the subrepresentations of the 3-irrep, given in table ???. We can show the qualitative difference

in the subspace dynamics by plotting the Liouville representation of the twirled *common noise*, and the *ZZ*-type noise. Specifically we plot $\tilde{\Lambda}^m$, where m is the sequence length. Figure ?? plots the maps in the Schur basis ordered so isotypic representations are adjacent to one another. Orange blocks have entries of 1, light green blocks are 0, and red blocks represent some decay parameters dependent on m . Panel (a) shows the fully diagonal *common noise* transfer matrix, where we can see the invariant trace subspace that was noted in the main text. Panel (b) shows the general case for twirled *ZZ*-type noise, given by equation A.24. The off-diagonal entries exactly couple the three subspaces of the 3-irrep. We note that the resulting block diagonal form is inconvenient to express as a general multi-exponential decay profile as given in the main text for *common* and local depolarising type noise. However, the resulting block diagonal Liouville form may be used directly with experimental data to fit these parameters of interest. Further, we may still explore the expected asymptotic decays due to this model, and find that under dominant *ZZ*-type noise, we expect an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/2$.

We may recover a simple diagonal Liouville form in the case of stochastic *ZZ* noise, given by,

$$\mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ}(Z \otimes Z)\rho(Z \otimes Z), \quad (\text{A.25})$$

where p_{ZZ} is the probability of a correlated phase flip error on both qubits. In the irrep basis, the following sector decays are found for the stochastic *ZZ* error,

$$\begin{aligned} \alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} &= 1 - 4p_{ZZ}/3, \\ \alpha_{tr} = \alpha_{dt} &= 1. \end{aligned} \quad (\text{A.26})$$

We show this dependence in figure A.1 (c). The same intermediate asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 0.5$ is expected.

The difference in correlated decay structure due to *common*, depolarising, and *ZZ*-type errors suggests that twirling over $Cl_1^{\otimes 2}$ both simplifies a generic error map to be measurable via RB, but also, retains some noise-dependent structure

of the correlated decay profiles. Designing optimal initial and final states, as well as optimal sampling strategies for extracting a larger set of error parameters covering multiple physical error mechanisms is left for future work. A possible strategy to avoid introducing arbitrary error channels, is to be agnostic of the physical error mechanisms, and rather directly extract the $\vec{\alpha}$ and $\vec{m}$ parameters described in section ???. Analogous protocols to this is described by character RB [70], and tomography via RB [71].

A.8 Magnus expansion

The unitary evolution due to a time dependent Hamiltonian may be written in the ordered, short-time limit as,

$$U(t) = \lim_{N \rightarrow \infty} e^{\frac{-i}{\hbar} \hat{H}(t_N) \Delta t} \dots e^{\frac{-i}{\hbar} \hat{H}(t_2) \Delta t} e^{\frac{-i}{\hbar} \hat{H}(t_1) \Delta t} = e^{A(t)}, \quad (\text{A.27})$$

where $\Delta t = t/N$, and $t_n = n\Delta t$, and $A(t)$ is a time dependent exponent, which is expanded in the Magnus construction to [34],

$$A(t) = \sum_{k=1}^{\infty} A_k(t). \quad (\text{A.28})$$

The first three $A_k(t)$ terms are given by,

$$A_1(t) = \frac{-i}{\hbar} \int_0^t dt_1 \hat{H}(t_1) \quad (\text{A.29})$$

$$A_2(t) = -\frac{1}{2\hbar^2} \int_0^t \int_0^{t_1} dt_1 dt_2 [\hat{H}(t_1), \hat{H}(t_2)] \quad (\text{A.30})$$

$$A_3(t) = \frac{i}{6\hbar^3} \int_0^t \int_0^{t_1} \int_0^{t_2} dt_1 dt_2 dt_3 \left([\hat{H}(t_1), [\hat{H}(t_2), \hat{H}(t_3)]] + [\hat{H}(t_3), [\hat{H}(t_2), \hat{H}(t_1)]] \right), \quad (\text{A.31})$$

where we can see that if the Hamiltonian fails to commute with itself at different times, the higher-order Magnus terms become relevant. This may be understood as the continuous analogy of the Baker-Campbell-Hausdorff formula. It is particularly useful in this thesis to understand the origin of the geometric phase in the MS gate scheme, and the non-linear bosonic interactions generated in the “two-SDF” protocol given in § 1.1.1.

A.8.1 “Two-SDF” dynamics

Here we analyse the “two-SDF” Hamiltonian given in equation 1.5 using the Magnus expansion. We recover effective Hamiltonians for both Gaussian interactions, as well as for tri-squeezing.

The “two-SDF” Hamiltonian is reproduced for convenience here,

$$\hat{H}_{\text{2SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (\text{A.32})$$

$$= \frac{\hbar\Omega_\alpha}{2}\hat{\sigma}_\alpha \left(\hat{a}_\alpha e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}_\alpha^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (\text{A.33})$$

$$+ \frac{\hbar\Omega_\beta}{2}\hat{\sigma}_\beta \left(\hat{a}_\beta e^{i(\Delta_\beta t + \phi_\beta)} + \hat{a}_\beta^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right). \quad (\text{A.34})$$

The first three terms of the corresponding Magnus expansion are found by inserting $\hat{H}_{\text{2SDF}}$ into equations A.29. We shall later rationalise that with certain values of SDF magnitudes, detunings, and phases, we may arbitrarily suppress the higher order terms.

The A_1 term appears as separable fast oscillations in phase space due to each SDF,

$$A_1(t) = -i \sum_{i \in \{\alpha, \beta\}} \left(\frac{\Omega_i \hat{\sigma}_i}{\Delta_i} \sin \left(\frac{\Delta_i t}{2} \right) \left(\hat{a}_i e^{i\left(\frac{\Delta_i t}{2} + \phi_i\right)} + \hat{a}_i^\dagger e^{-i\left(\frac{\Delta_i t}{2} + \phi_i\right)} \right) \right). \quad (\text{A.35})$$

This is effectively the first-order only dynamics of the displacement operator leading to excursions in phase-space. The effects of this term may be entirely suppressed at certain gate durations, $A_1(T) = 0$, where T is an integer multiple of both $\tau_\alpha = 2\pi/\Delta_\alpha$ and $\tau_\beta = 2\pi/\Delta_\beta$. Excursions may also be made arbitrarily small by ensuring $\Omega_i \ll \Delta_i$.

The commutator in the A_2 term may be expanded to reveal four relevant components,

$$A_2(t) = S_\alpha(t) + S_\beta(t) + X_{\text{pair}}(t) + X_{\text{exch}}(t), \quad (\text{A.36})$$

where S denotes “self-terms”, and X “cross-terms” of the two simultaneously applied SDFs. We label the first cross-term “pair” due to the pair creation and annihilation of excitations, and the second group, “exch” is labelled due to driving the exchange of excitation.

The self-terms, with the form,

$$S_i(t) = \frac{i\Omega_i^2}{4} \left(\frac{\sin(\Delta_i t)}{\Delta_i^2} - \frac{t}{\Delta_i} \right) \quad (\text{A.37})$$

are recognisable from section ?? as geometric phase factors. Taking the assumption that $t \gg \frac{1}{\Delta_i}$, we get a linearly growing geometric phase term, $S_i(t) \approx \frac{-i\Omega_i^2 t}{4\Delta_i}$. In the case that this geometric phase acts globally on the system, they may be ignored. If the accumulated phase is spin, or oscillator state dependent, then interesting dynamics may be observed, however these effects are out of the scope of this thesis. The remaining cross-terms contain the 2nd order non-linear bosonic interactions we are interested in emulating. We set the two detunings relative to one another by, $\Delta_\beta = m\Delta_\alpha$. Expanding the commutator of the cross terms we find for general m the following interactions, grouped by their effect on the oscillator state,

$$X_{\text{pair}} = \frac{i\epsilon_{\alpha\beta\gamma}\Omega_\alpha\Omega_\beta}{4\Delta}\hat{\sigma}_\gamma\left(\hat{a}_\alpha\hat{a}_\beta e^{i\phi_p}J(m,t) + \hat{a}_\alpha^\dagger\hat{a}_\beta^\dagger e^{-i\phi_p}J(m,t)^\dagger\right), \quad (\text{A.38})$$

$$X_{\text{exch}} = \frac{i\epsilon_{\alpha\beta\gamma}\Omega_\alpha\Omega_\beta}{8\Delta}\left(\{\hat{a}_\alpha, \hat{a}_\beta^\dagger\}e^{i\phi_m}F(m,t) + \{\hat{a}_\alpha^\dagger, \hat{a}_\beta\}e^{-i\phi_m}F(m,t)^\dagger\right), \quad (\text{A.39})$$

with $\phi_p = \phi_\alpha + \phi_\beta$, and $\phi_m = \phi_\alpha - \phi_\beta$, and J and F containing resonance like terms,

$$J(m,t) = \frac{1-m}{m}\left(\frac{e^{i(m+1)\Delta t} - 1}{(m+1)\Delta t}\right)t, \quad (\text{A.40})$$

$$F(m,t) = \left(\frac{m+1}{m}\frac{e^{-i(m-1)\Delta t} - 1}{(m-1)\Delta t} + \frac{e^{i\Delta t} - e^{-im\Delta t}}{m\Delta t}\right)t. \quad (\text{A.41})$$

We now may find the effective Hamiltonians governing the dynamic evolution for certain resonance conditions. We can convert from the generator form, $A(T)$, to familiar effective time-independent Hamiltonian, $\hat{H}_{\text{eff}}$ for a given time, T , by $\hat{H}_{\text{eff}} = \frac{i\hbar}{T}A(T)$.

Taking the limit, $m = -1$, we find $F(-1,t) = 0$ and $J(-1,t) = -2it$, leading to a time-linear term X_{pair} , with effective Hamiltonian,

$$\hat{H}_{\text{eff,pair}} = \frac{\epsilon_{\alpha\beta\gamma}\hbar\Omega_2}{2}\hat{\sigma}_\gamma\left(\hat{a}_\alpha\hat{a}_\beta e^{i\phi'_p} + \hat{a}_\alpha^\dagger\hat{a}_\beta^\dagger e^{-i\phi'_p}\right), \quad (\text{A.42})$$

where $\Omega_2 = \frac{\Omega_\alpha\Omega_\beta}{\Delta}$, and $\phi'_p = \phi_\alpha + \phi_\beta + \frac{\pi}{2}$. For $a_\alpha^\dagger = a_\beta^\dagger$ this is the squeezing interaction, whereas for $a_\alpha^\dagger \neq a_\beta^\dagger$ this is two-mode squeezing.

Taking the limit, $m = +1$, we find $J(1,t) = 0$ and $F(1,t) = -2it(1 - \text{sinc}(\Delta t))$.

If we consider the secular term only, we find the time-linear term X_{exch} . Now

considering the anticommutation relationships between ladder operators we find for $a_\alpha^\dagger = a_\beta^\dagger$, an effective Hamiltonian,

$$\hat{H}_{\text{eff,parity}} = \frac{\varepsilon_{\alpha\beta\gamma}\hbar\Omega_2}{2}\hat{\sigma}_\gamma \sin(\phi_m)(2\hat{a}_\alpha^\dagger\hat{a}_\alpha + 1), \quad (\text{A.43})$$

which is the parity, or in this case “dispersive” Hamiltonian due to the spin-conditioning.

Finally, considering $a_\alpha^\dagger \neq a_\beta^\dagger$, we recover the beamsplitter interaction, with effective Hamiltonian,

$$\hat{H}_{\text{eff,bs}} = \frac{\varepsilon_{\alpha\beta\gamma}\hbar\Omega_2}{2}\hat{\sigma}_\gamma \left(\hat{a}_\alpha\hat{a}_\beta^\dagger e^{i\phi'_m} + \hat{a}_\alpha^\dagger\hat{a}_\beta e^{-i\phi'_m} \right), \quad (\text{A.44})$$

with $\phi'_m = \phi_\alpha - \phi_\beta + \frac{\pi}{2}$.

Similar arguments hold for the higher order terms, where, for example, we recover the tri-squeezing interaction for $m = -2$, with effective Hamiltonian of the form,

$$\hat{H}_{\text{eff,tri}} = \frac{\hbar\Omega_3}{2}\hat{\sigma}_\beta \left(\hat{a}_\alpha^3 e^{i\phi} + \hat{a}_\alpha^{\dagger 3} e^{-i\phi} \right), \quad (\text{A.45})$$

with $\Omega_3 = \frac{\Omega_\alpha^2\Omega_\beta}{2\Delta^2}$.

For further details on the use of these higher order effective interactions refer to the following two theses:[3, 4].

A.9 Derangement for $k = 2$

Huggins [8] provides a proof of the identity given in equation 1.21, which I reproduce here for the simple example of $k = 2$. The derangement operator for two copies is simply the swap operator, $S^{(2)} = \text{SWAP}$, and we will assume the observable acts only on the first copy, giving $\hat{O}^1 = \hat{O} \otimes \mathbb{I}$. The identity is then,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (\text{A.46})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \langle i_1 | \hat{O} | k_1 \rangle \delta_{i_2, k_2} \langle k_1, k_2 | S^{(2)} | j_1, j_2 \rangle \langle j_1 | \rho | i_1 \rangle \langle j_2 | \rho | i_2 \rangle \quad (\text{A.47})$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \hat{O}_{i_1, k_1} \delta_{i_2, k_2} \delta_{k_1, j_2} \delta_{k_2, j_1} \rho_{j_1, i_1} \rho_{j_2, i_2} \quad (\text{A.48})$$

and contracting the Kronecker deltas, we find,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \tag{A.49}$$

$$= \sum_{i_1, j_1, j_1} \hat{O}_{i_1, j_2} \rho_{j_1, i_1} \rho_{j_2, j_1} \tag{A.50}$$

$$= \text{Tr}[\hat{O} \rho^2], \tag{A.51}$$

with the last equality following from the definition of matrix multiplication.

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